Research Methodology
Course Progress 44.72% (11 of 12)
DOE Part 1
DOE part 2
DOE part 3
DOE part 4
DOE part 5
Quiz: Week 7 : Assignment
Week 7 Feedback Form : Research Methodology
Week 8- Department specific discussions
Week 9
Week 10
Week 11
Week 12
Text Transcripts
Books
Download Videos

Assessment

Week 7 : Assignment

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-09-01, 20:11 IST.

Due date on 2026-09-09, 23:59 IST

1. In the true experimental research design: 1 Point

- No control on external variables
- Less control on external variables
- (•) Control on external variables by randomization
- No use of randomization

Yes, the answer is correct. Score: 1 | Accepted answer is Control on external variables by randomization

2. Which of the following is an incomplete experimental research design? 1 Point

- Before and after with control group design
- Factorial design
- (•) One group before-after design
- Latin Square design

Yes, the answer is correct. Score: 1 | Accepted answer is One group before-after design

3. Coefficient of Determination is said to be unity when - 1 Point

- [x] Error Sum of Squares is zero
- [ ] Total Sum of Squares is zero
- [ ] Regression Sum of Squares is zero
- [x] Regression Sum of Squares is same as Total Sum of Squares

Yes, the answer is correct. Score: 1 | Accepted answer is Error Sum of Squares is zero, Regression Sum of Squares is same as Total Sum of Squares

4. The probability density function describes the following - 1 Point

- Probability of a phenomenon occurring in macro scale
- Probability of a phenomenon occurring in micro scale
- (•) Distribution of probabilities in the continuous random variable domain
- Distribution of probabilities in some particular set of undefined variables

Yes, the answer is correct. Score: 1 | Accepted answer is Distribution of probabilities in the continuous random variable domain

5. During Response Surface Methodology's Canonical Analysis involving two factors, maximum response is said to be achieved when the eigen values of the B matrix are 1 Point

- all positive
- (•) all negative
- all zero
- such that one is positive and the other is negative

Yes, the answer is correct. Score: 1 | Accepted answer is all negative

6. Experiments are conducted in laboratories to 1 Point

- Verify scientific predictions
- Obtain trends when theory is not known
- Find effect of different variables on the desired response
- (•) All of the above

Yes, the answer is correct. Score: 1 | Accepted answer is All of the above

7. SPC and Design of Experiments are very closely interrelated. 1 Point

- (•) True
- False

Yes, the answer is correct. Score: 1 | Accepted answer is True

8. Which of these is not an application of experimental design in the field of Engineering design? 1 Point

- Evaluation and comparison of basic design configurations
- Evaluation of material alternatives
- Evaluation of key product design parameters that impact performance
- (selected) Evaluation of correct process to develop the product

Yes, the answer is correct. Score: 1

Accepted answer is Evaluation of correct process to develop the product

9. The average estimated from a data collection is 50 cm. The quantity (quantities) which has the same units as the data is (are) 1 Point

- Standard deviation
- Median
- Mode
- (selected) All of the above

Yes, the answer is correct. Score: 1

Accepted answer is All of the above

10. Sample collected from an instrument giving binary data has the following values {1, -1, -1, 1, 1, -1} Mean of the above data set is 1 Point

- -1
- (selected) 0
- 1
- 3

Yes, the answer is correct. Score: 1

Accepted answer is 0